

Worachit Ketrungsri

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WORK EXPERIENCE

French National Centre for Scientific Research (CNRS)

France, Toulon

Data Scientist (Internship)

Jun, 2023 – Jul, 2023

- Improved the object detection model for elongated structures in low surface brightness (LSB) galaxy images, increasing AP50 score from 0 to 0.45.

Vidyasirimedhi Institute of Science and Technology

Thailand, Bangkok

Software Engineer

Jan, 2022 – Sep, 2022

- Developed IoT solutions in C# for integrating commercial devices in the lab, such as exoskeletons, motion capture systems, and EMG, using the MQTT protocol and devices API (Paper published in IEEE).
- Designed and implemented a GUI in PyQt for gait experiments and developed a data collection pipeline, reducing the data collection time by 30%.

Robolab OSKN

Thailand, Bangkok

Software Engineer

Mar, 2021 – Oct, 2021

- Collaborated with a team to develop a web-based block coding platform for kids, which supports code uploading to the differential drive robots developed by our company. Tested the product with over 80 middle school students, receiving a satisfactory score of 3.8/5.

COXSYS Robotics

Thailand, Bangkok

Software Engineer (Internship)

Nov, 2020 – Jan, 2021

- Developed a Python-based path-finding algorithm utilizing Voronoi diagrams generated from building layouts and built a .deb package from the Python script.

PROJECT WORK

RAGHack

Sep 2024

- Participated in Microsoft's "RAGHack" competition, developing a Retrieval-Augmented Generation (RAG) system to perform semantic search on a book database stored in PostgreSQL.

Stock Price Prediction

Thailand, Bangkok

Nov, 2022 – Feb, 2023

- Developed a stock price prediction model using LSTM, achieving a Root Mean Squared Percentage Error (RMSPE) of 12% on US stocks such as AMZN and GOOGL
- Simulated the stochastic volatility model of GSPC's option price in Python using the Heston Model.

Oceanographic data analysis

France, Toulon

Sep, 2022 – Nov, 2022

- Contributed to a team responsible for deploying oceanographic instruments and analyzing data on ocean circulation and wind direction in the Mediterranean Sea using Python, utilizing over 20 drifters and CTD devices for data collection.

ZEBRA: A Robot Manipulator for Waste Sorting

Thailand, Bangkok

Mar, 2020 – Aug, 2020

- Collaborated with a team to design the instance segmentation (Mask RCNN) in TensorFlow. The model can identify and accurately segment cans, snack bags, and bottles with an mAP score of 0.905.

Jaundice Disease Classification

Thailand, Bangkok

Nov, 2019 – May, 2020

- Contributed to a team that designed a machine learning-based pre-diagnostic classification algorithm.
- Developed the eye segmentation algorithm using U-net architecture

SAM: A Development of Intelligent Medical Platform

Thailand, Bangkok

Jan, 2019 – May, 2019

- Developed a Melanoma disease classification website using the Django framework and MongoDB as a database. The platform provides pre-diagnostic results from skin images uploaded by users.
- Deployed and integrated a machine learning model as a service on Google Kubernetes Engine (GKE). The microservice communicates with the website via RESTful API, achieving an average response time of under 800 milliseconds per API call with a CPU cluster. The project was selected to present at the 2nd World Congress on Undergraduate Research

EDUCATION

Jaume I University (UJI)

M.Sc. in Marine and Maritime Intelligent Robotics (MIR) - Erasmus+ scholarship (Second year)

Spain, Castellón de la Plana

Aug 2024

Université de Toulon (UTLN)

M.Sc. in Marine and Maritime Intelligent Robotics (MIR) - Erasmus+ scholarship (First year)

France, Toulon

GPA: 16.04/20.0 (First class honor)

Jul 2023

King Mongkut's university of technology Thonburi (KMUTT)

B.E. in Robotics and Automation

Thailand, Bangkok

GPA: 3.77/4.0 (First class honor)

Jul 2021

EXTRA SECTION

- First Thai student to receive the ERASMUS+ scholarship for the Marine Intelligent Robotics (MIR) program (2021)
- Reputation Award for contributions that built the university's reputation (2019)
- Participated as an Undergraduate Researcher in the World Congress on Undergraduate Research (World CUR) (2019)
- Best Design award in the Smart Agricultural Robot Contest (2018)
- Advanced to the first round of the Computer Olympiad (POSN) for high school students (2016)

OTHER SKILLS

Language: Thai (Native), English (IELTS 7.0)

Programming: Python (familiar), C (familiar), MATLAB (familiar), C++, C# Ladder, SQL

OS: Window, Linux Ubuntu

Other: GIT, Docker, Machine learning, JSON

Interests: Software Development, Robotics, Machine learning, Embedded System, Quantitative trading